Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/16/2006
PI Name:	Stebe, Kathleen J. Ph.D.		
Project Title:	Using Surfactants to Control Bubble Growth Coalescence in Nucleate Pool Boiling		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kjs@jhu.edu	**Fax:**	FY 410 516-5510
PI Organization Type:	UNIVERSITY	**Phone:**	410 516-7769
Organization Name:	Johns Hopkins University		
PI Address 1:	Department of Chemical and Biomolecular Engineering		
PI Address 2:	3400 N. Charles Street		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21218-2694	**Congressional District:**	7
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	06/01/2000	**End Date:**	05/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Kizito, John P	**Contact Phone:**	216-433-2275
Contact Email:	John.P.Kizito@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Balasubramaniam, R. (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-812		
Performance Goal No.:			
Performance Goal Text:			

Drops can become elongated under gravity or other applied fields, and subsequently pinch-off by the action of surface tension. Surfactants are molecules that spontaneously adsorb at fluid interfaces (for example, between a drop and a surrounding fluid) and lower the surface tension, and hence decrease the work required to form new interface. If surfactants collect any location on an interface, they can drop the surface tension, causing drops to deform into shapes they would not assume in the absence of surfactants. The aim of this research was to understand how surfactants can alter drop and bubble evolution and detachment with the intent to exploit this understanding in heat transfer applications. Extensions of this research could be used to study multiple droplets and the role of surfactants in reducing drop coalescence.
Surfactant distribution on an interface is governed by the rates of surface expansion or contraction and the rates of surfactant mass transfe. When surfaces contract faster than surfactants can desorb or diffuse away, surfactants will collect above their equilibrium packing. When surfaces expand faster than surfactants can adsorb or diffuse, the surface

Task Description:	concentration will be reduced. For complex motions like those encountered in the evolution of a drop from formation to break-off, some regions of the interface expand while others contract, so drops can have dilute interfaces (and high surface tensions) in some regions coexisting with concentrated interfaces (and low surface tensions) in others. Stress jumps at the interface are balanced tangentially by gradients in tension, and normally by the product of surface tension and curvature. Thus, complex surfactant distributions alter the drop mechanics. To understand this behavior, we have studied three model problems numerically, two of which are completed, and third of which is ongoing. In all three, an isothermal system was considered. (i) For a quiescent system, how do surfactant adsorption/desorption kinetics compete with surfactant bulk diffusion kinetics? In particular, how does the prevailing mass transfer regime depend on the drop radius? In this work, a new characteristic length scale RDK is identified which depends only upon surfactant physicochemical parameters, and is independent of surfactant concentration. For drop radii smaller than RDK, mass transfer is kinetically controlled. For drop radii than RDK, the mass transfer is diffusion controlled. (ii) In the adsorption/desorption regime, how do surfactants alter the evolution of an injected drop studied from an initially planar interface to detachment? Surfactants are shown to change drop behavior significantly. Phase diagrams of drop breakup behavior are developed as a function of surfactant concentration and adsorption/desorption kinetics. (iii) For an injected drop, how do surfactant adsorption/desorption kinetics compete with surfactant bulk diffusion kinetics? We hypothesize that RDK will play a significant role. Scaling arguments suggest that for diffusion boundary layer thicknesses around the drop smaller than RDK, mass transfer will be kinetically controlled, whereas for diffusion boundary layer thicknesses greater than RDK, the mass transfer will be diffusion controlled. This aspect of the work is continuing. Summary In the originally proposed research, we undertook to understand the effects of surfactants on drop evolution and detachment of single drops, pairs of drops, and in boiling heat transfer. The work on pairs of drops was not pursued at JHU because of budget revisions upon funding. Some heat transfer experiments were pursued at NASA. Three related numerical problems have been studied at JHU. The first two are complete. The last is ongoing. To elucidate the manner in which surfactants create new break off modes with an aim to harnessing these new mechanisms in applications such as heat transfer in microgravity, drop and bubble management in microfluidics. Furthermore, these studies can be exploited in the characterization of surfactant dynamics. Buoyancy plays an important role in nucleate pool boiling on the Earth. When the heater surface is below the liquid, vapor bubbles depart from the surface due to the action of buoyancy. This promotes the rate of heat transfer from the heated surface. When the heating surface is above the liquid under normal gravity, the vapor bubbles do not move away from the surface as buoyancy pushes the bubbles toward the heater. The growing bubbles soon coalesce and a regime of film boiling is achieved. In such a mode of operation the benefits of nucleate boiling are not utilized. For efficient operation of the system in the regime of nucleate boiling, the dynamics of the vapor bubbles must be understood and controlled. This is even more important in reduced gravity because in the absence of buoyancy, the primary mechanism for the removal of bubbles from the heating surface is eliminated. We propose to control the dynamics of bubbles in nucleate boiling in reduced gravity by the addition of soluble surfactants to the liquid.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Drop growth and detachment studies are currently used to infer surfactant properties such as adsorption-desorption rates or diffusivity in experiments like the drop weight technique or maximum bubble pressure techniques. The results from our studies will allow these methods to be made more robust, in that incorrect, ad hoc assumptions about drop shapes, growth modes and detachment mechansims can be eliminated. In microgravity experiments in which immiscible fluids are used to make discrete fluid elements (e.g. drop or bubble formation in microfluidics channels surrounded by other fluids) surfactants have been deminstrated in experiment to strongly alter drop formation and detachment mechanisms, without a detailed understanding of how surfactant physical chemistry dictates the observed changes. Such an understanding will be advanced by this research. In the creation of dispersions (drops, emulsions, bubbles, sprays) surfactants play a strong role in altering drop and bubble size, stability, etc,. These fields will be impacted by the results of this current study.
Task Progress:	To understand this behavior, we have studied three model problems numerically, two of which are completed, and third of which is ongoing. In all three, an isothermal system was considered. (i) For a quiescent system, how do surfactant adsorption/desorption kinetics compete with surfactant bulk diffusion kinetics? In particular, how does the prevailing mass transfer regime depend on the drop radius? In this work, a new characteristic length scale RDK is identified which depends only upon surfactant physicochemical parameters, and is independent of surfactant concentration. For drop radii smaller than RDK, mass transfer is kinetically controlled. For drop radii than RDK, the mass transfer is diffusion controlled. (ii) In the adsorption/desorption regime, how do surfactants alter the evolution of an injected drop studied from an initially planar interface to detachment? Surfactants are shown to change drop behavior significantly. Phase diagrams of drop breakup behavior are developed as a function of surfactant concentration and adsorption/desorption kinetics. (iii) For an injected drop, how do surfactant adsorption/desorption kinetics compete with surfactant bulk diffusion kinetics? We hypothesize that RDK will play a significant role. Scaling arguments suggest that for diffusion boundary layer thicknesses around the drop smaller than RDK, mass transfer will be kinetically controlled, whereas for diffusion boundary layer thicknesses greater than RDK, the mass transfer will be diffusion controlled. This aspect of the work is continuing.
Bibliography Type:	Description: (Last Updated: 01/16/2006)
Articles in Peer-reviewed Journals	Jin, F., Gupta, N., and Stebe, K.J. "“The detachment of a viscous drop in a viscous solution in the presence of a soluble surfactant” " Phys. Fluids , Feb-2006

Articles in Peer-reviewed Journals	Jin, F., Balasubramiam, R. and Stebe, K.J. ""Surfactant Adsorption to Spherical Particles: The Intrinsic Length Scale Governing the Shift from Diffusion to Kinetic Controlled Mass Transfer" " J. of Adhesion, 80, 773-796, 2004 , Jan-2004